MUHAMAD AIMAN FAIZ

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Portfolio: muhamadaimanfaiz.github.io/Mechanical-Engineering

PROFILE

MEng Mechanical Engineering graduate from UCL (2:1) with hands-on experience in structural analysis, FEA, CFD, experimental testing and design optimisation, gained through an industry-linked final-year project with Cannondale, an individual water treatment system design and a battery tray redesign that reduced mass by 30.1%. Experienced in coordinating contractors and technicians across 60+ residential properties while delivering to competing deadlines. Seeking a graduate mechanical, design or structural engineering role in London.

EDUCATION

University College London (UCL), London, United Kingdom | Sep 2022 – Sep 2026
MEng Mechanical Engineering, Upper Second-Class Honours (2:1)

- Modules: Solid Mechanics, Elasticity and Plasticity, Dynamics and Control, Advanced Thermodynamics and Fluid Mechanics, Mathematical Modelling and Analysis II, Project Management
- MARA Young Talent Programme (YTP) Scholar

Kolej MARA Seremban, Malaysia | Jun 2020 – Jun 2022

- Cambridge A-Levels: Mathematics (A*), Physics (A), Chemistry (A*), 15/15 points. Gold Award, Varsity Maths 2021. IELTS 7.5

ENGINEERING PROJECTS

Mountain Bike Frame Optimisation | MEng Final-Year Project, with Cannondale | Sep 2025 – Jun 2026

- Contributed to the design, manufacture and validation of a universal test rig in Fusion 360, with a front gimbal and rear hub replicating the FE boundary conditions, and verified its rigidity with FEA at 2.5 times the planned test loads (minimum safety factor 2).
- Developed an experimental measurement system using 16 strain gauges in eight back-to-back pairs, Wheatstone bridge circuitry and HX711 modules with an Arduino, validated against beam theory with under 1% axial and approximately 0.83% bending difference.
- Combined strain gauge and dial gauge displacement data across three load cases of up to 90 kg with Abaqus FEA outputs to validate analytical predictions and identify differences between modelled and observed performance.
- Evaluated a laser deflectometer against a dial gauge for displacement measurement, validating the deflectometer on a cantilever beam against FEA, then selected the dial gauge with a custom-designed 3D-printed adapter for repeatable readings on the full-frame rig.
- Secured Cannondale as the project's industry partner after approaching several bicycle manufacturers, obtaining a Trail 2 mountain bike, a test frame and the manufacturer's 3D CAD geometry used to build the finite element model.

Water Clean-Up Solution Design | UCL Individual Project, Pigeon Creek, Indiana | Sep 2024 – Jun 2025

- Designed a modular water treatment system to divert and treat 450 L/s (20% of creek flow) for reuse by a nearby metal fabrication plant, covering trash interception, flocculation, sedimentation, filtration, UV disinfection and storage.
- Sized each treatment stage from its retention time with a 1.3 safety factor, and used Bernoulli and Darcy–Weisbach head-loss calculations to size the transfer pumps.
- Selected reinforced concrete for the sedimentation basin in Granta EduPack and modelled the basin wall in Abaqus (2D plane strain, 3 m of water and 1 m of sludge), giving a peak von Mises stress of 27 MPa against a 40 MPa compressive strength.

Forensic Engineering: Crane Failure Investigation | UCL Group Design Project | Oct 2023 – Dec 2023

- Reviewed 14 technical documents on a toppled factory crane to identify the most credible failure pathways.
- Identified M24 stud bolts fitted in place of the specified M30 (22.5% lower tensile strength) and inverted locking nuts, and showed that the combined crane, load and bolt preload of 188.4 kN exceeded the 158.75 kN thread-stripping capacity.
- Proposed a redesigned crane with a larger base and a more secure base-to-plinth connection, alongside corrective actions for installation and commissioning.

Swappable Battery Tray Design and Optimisation | UCL Design Project | Sep 2023 – Jan 2024

- Defined requirements for a swappable battery tray: operation up to 240°C, Young's modulus of at least 60 GPa, yield strength of at least 50 MPa, a five-year life and a safety factor of 3.
- Screened 15 candidate materials in Granta EduPack and ranked nine shortlisted options on strength-to-weight ratio, thermal performance, cost, corrosion resistance and recyclability, selecting an aluminium-silicon-carbide composite.
- Evaluated the tray under a 2.943 kN distributed battery load using analytical calculations and FEA, then applied topology optimisation and generative design to reduce mass from 198.5 kg to 137 kg (30.1% reduction) at an estimated material cost of £615.13.

Additional projects: combustion analysis of a hexanol/octane fuel blend (2024), magnetorheological pipe damper study (2025), ANSYS Fluent CFD of a three-outlet manifold (2025), MATLAB Q4 finite element code verified against ANSYS APDL (2026), centrifugal impeller blade design (2024), CNC toolpaths in Fusion 360 CAM (2024).

WORK EXPERIENCE

CNC Property Services, United Kingdom | Property Management Operations Intern | Oct 2025 – Sep 2026

- Coordinated maintenance and remedial works across 60+ residential properties, liaising with 5–10 contractors, technicians and tenants to schedule access, track progress and follow technical issues through to resolution.
- Maintained compliance and maintenance records covering EICR, EPC, CP12 and fire-door remedial works, working with contractors and technical specialists to communicate defects and monitor repairs.
- Managed competing operational priorities and high-season scheduling, coordinating workloads, appointments and stakeholder requirements while adapting plans as issues changed.

University College London (UCL), United Kingdom | Room Monitor Officer | Sep 2025 – Jun 2026

- Conducted daily technical inspections across 5–10 teaching rooms, verifying AV, projection and control systems before scheduled sessions, and logging and escalating faults to support teams within defined timeframes.

Makan Cafe, United Kingdom | Customer Service and Operations Associate | Oct 2023 – Dec 2024

- Delivered customer service in a fast-paced hospitality environment, coordinating with kitchen and floor teams to resolve operational issues during busy periods.

STEM Private Tutoring, Malaysia | Mathematics and Physics Tutor | Feb 2020 – Apr 2020

- Delivered tailored Mathematics and Physics tuition, translating technical concepts into clear explanations suited to different levels of understanding.

LEADERSHIP AND ACTIVITIES

KUGEM Volleyball Club, United Kingdom | Founder and President | Oct 2022 – Jun 2026

- Led and coordinated 20+ active members, managing training schedules, tournament logistics and team organisation, and guided the team to GoMammoth Open League 2023 Champions, runner-up at Kent Malaysian Games 2024 and bronze at Warwick SEA Games 2025 and Nottingham Malaysian Games 2025.

Greater London Authority and TalentCorp Malaysia, London | Event Photographer | Mar 2026

- Photographed the GLA Annual Staff Iftar and the TalentCorp MyHEART Networking Session in London.

High Commission of Malaysia, London | Event Coordinator and Sales Assistant | Aug 2024

- Supported event logistics and sales operations for the International London Malaysia Carnival, a large public event, responding to attendee enquiries and adapting to changing demand.

Cultural and Traditional Music Society, Malaysia | Founder and President | Jul 2020 – Jul 2021

- Founded and led the society, coordinating two performances for audiences of 450+ while managing an approximately RM3,600 budget, equipment procurement and external professional coaches.

SKILLS

- **CAD and design:** Fusion 360 (including CAM), SolidWorks, Granta EduPack
- **Analysis and simulation:** FEA, Abaqus, ANSYS Mechanical APDL, ANSYS Fluent (CFD), topology optimisation, generative design
- **Testing and instrumentation:** strain gauges, Wheatstone bridge circuits, HX711 and Arduino data acquisition, dial gauges
- **Programming and data:** Python, MATLAB, Microsoft Excel
- **Languages:** English, Malay (native), Mandarin (developing)